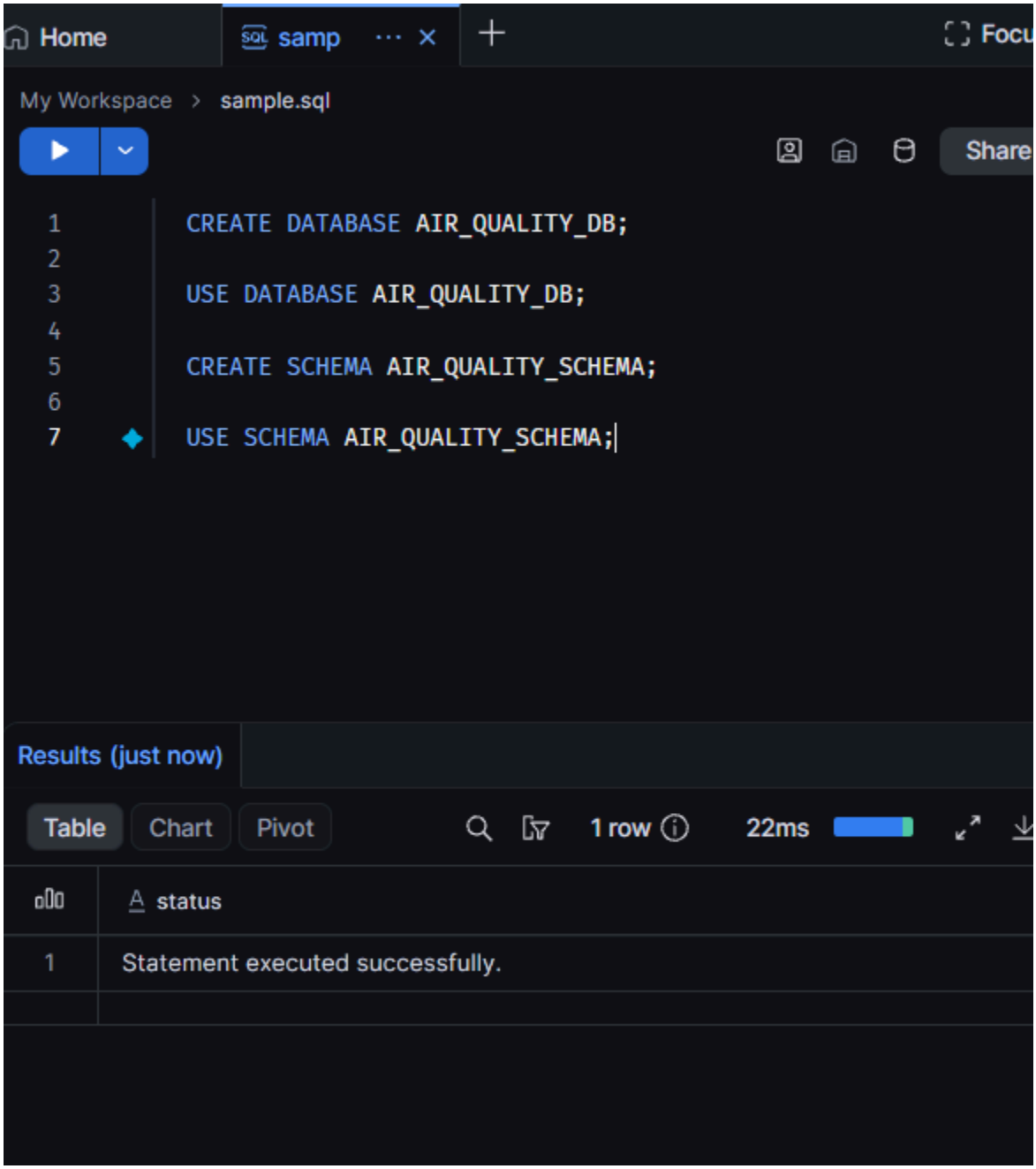
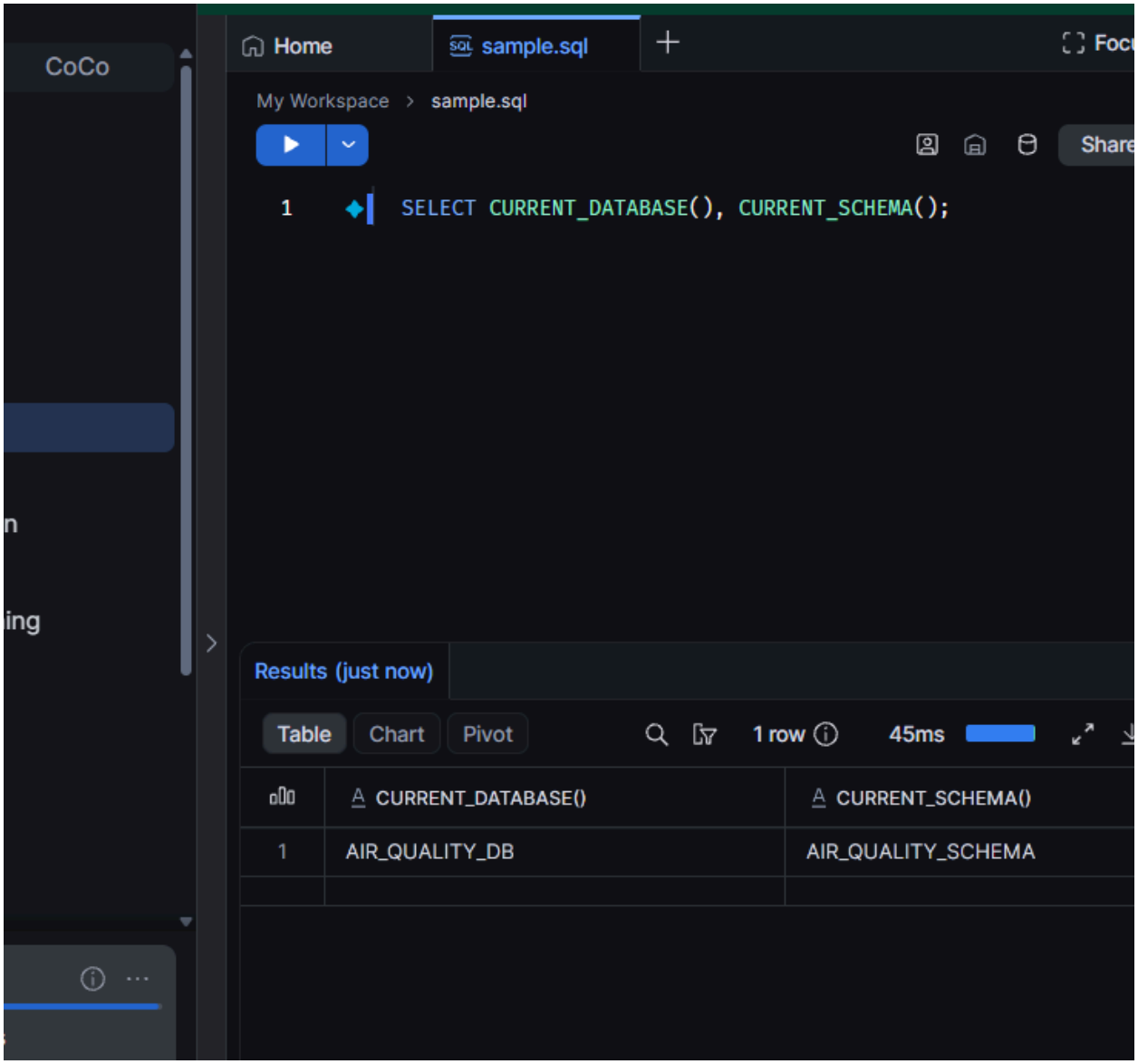
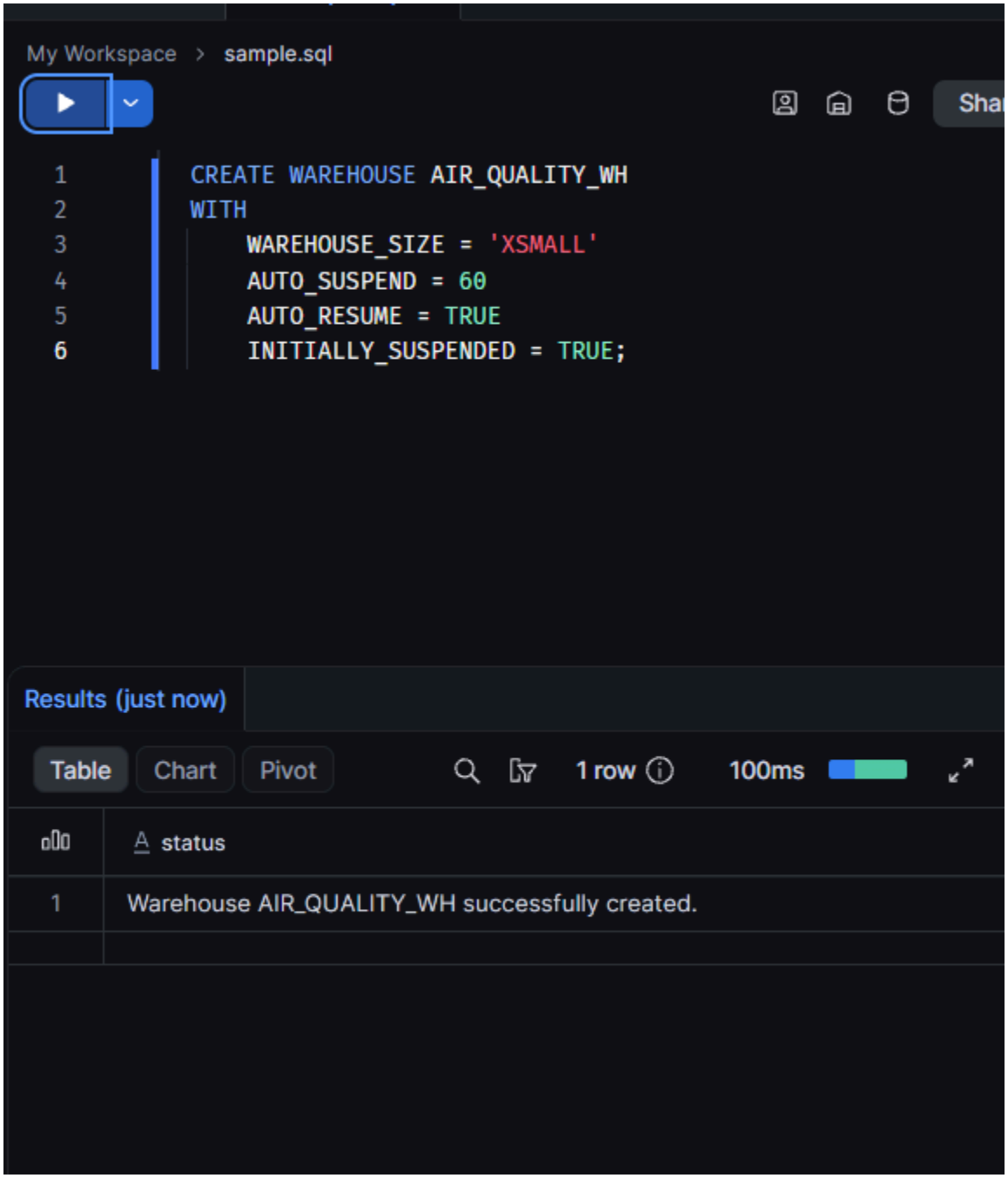
Create database and schema

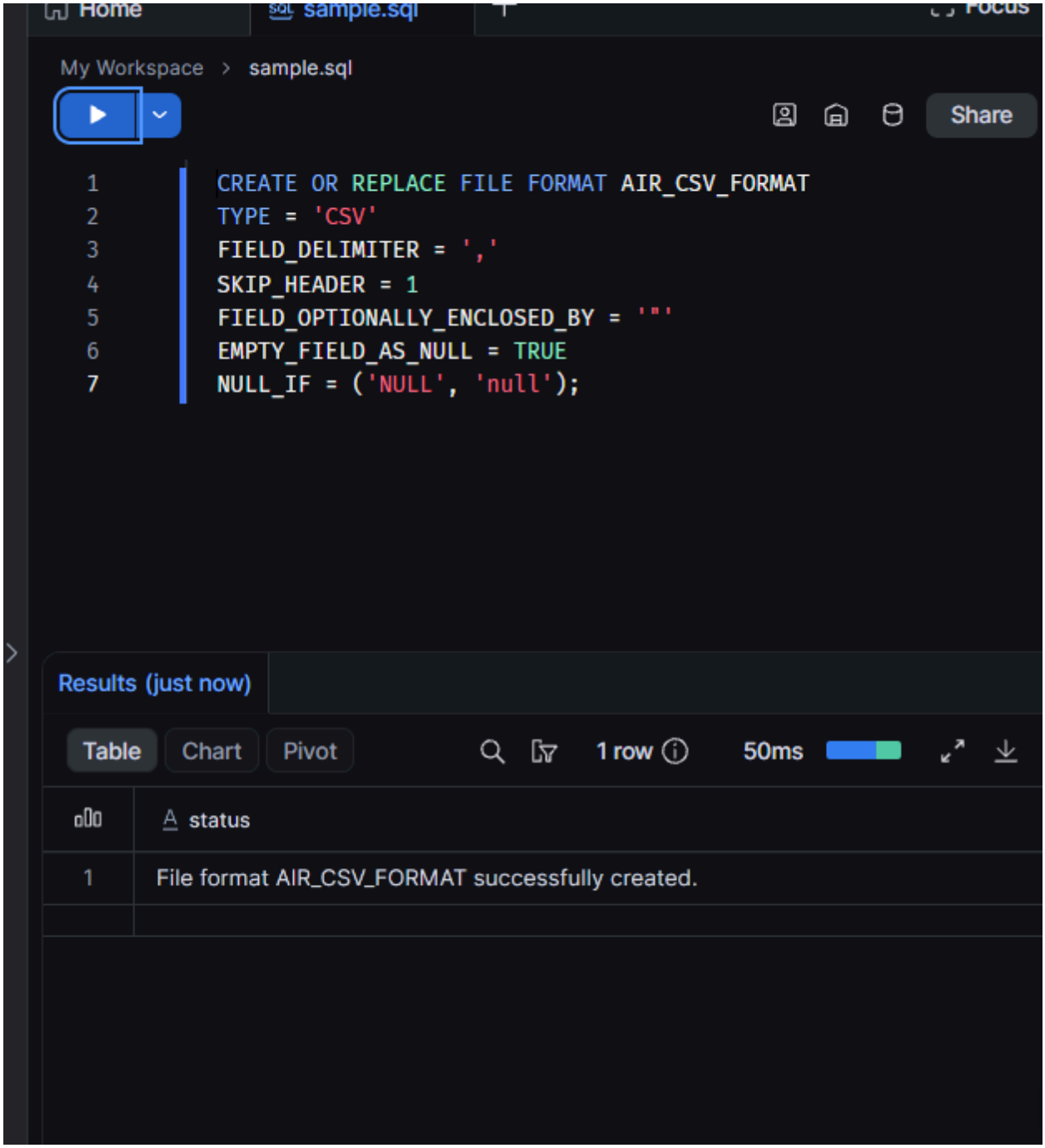


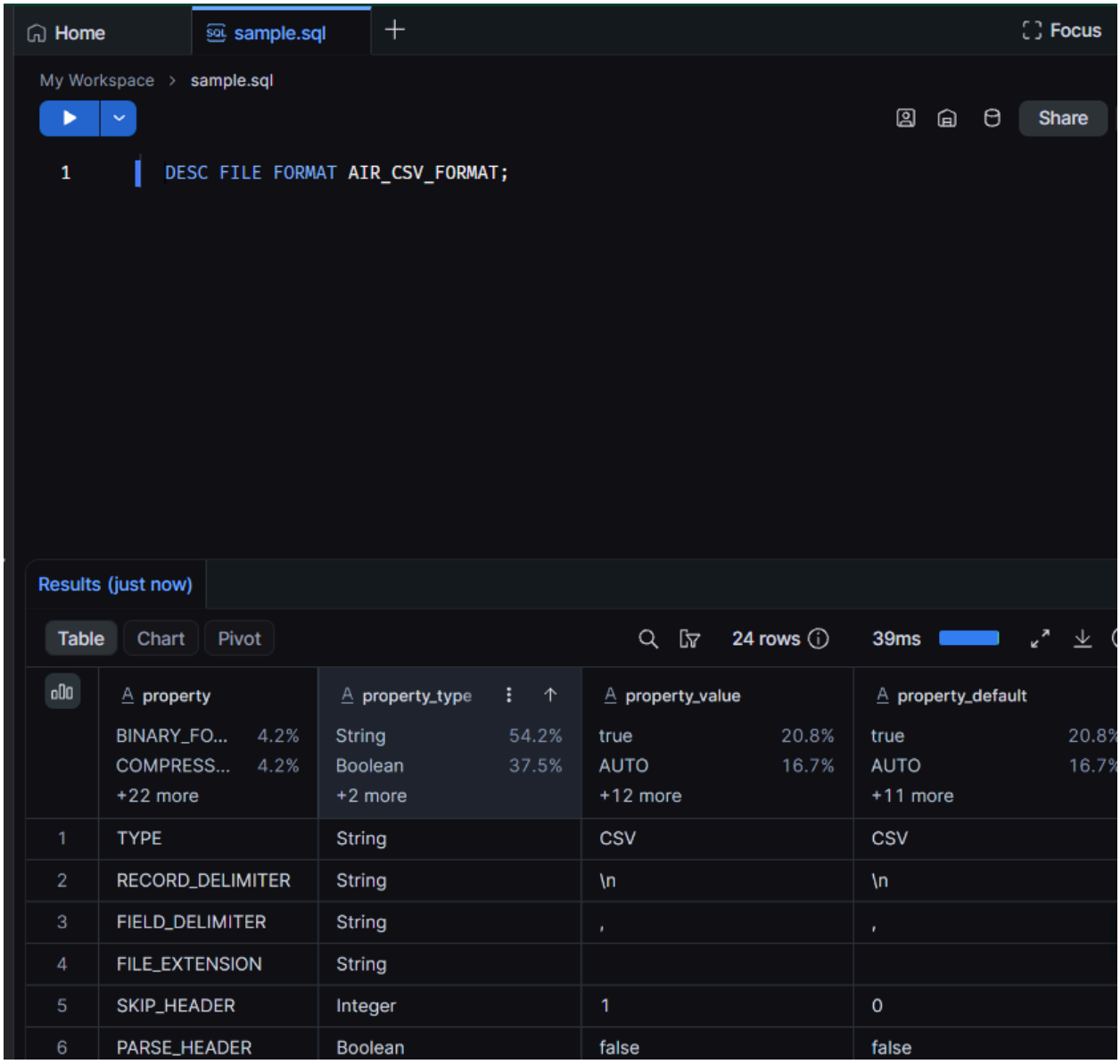

CoCo
Home
sample.sql
My Workspace
sample.sql
Share
1
SELECT CURRENT_DATABASE(), CURRENT_SCHEMA();
Results (just now)
Table
Chart
Pivot
1 row
45ms
CURRENT_DATABASE()
CURRENT_SCHEMA()
1
AIR_QUALITY_DB
AIR_QUALITY_SCHEMA

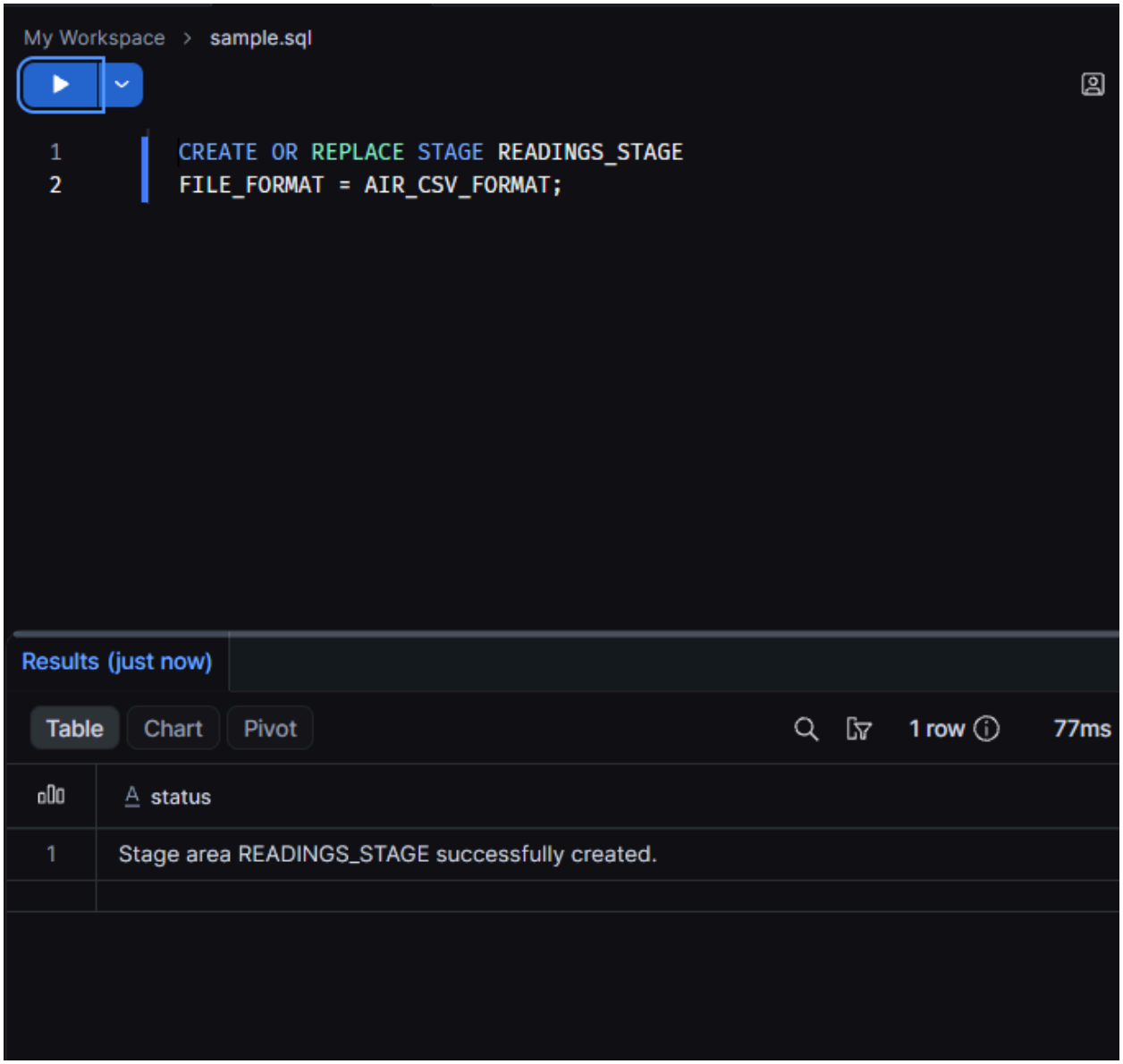
Create a warehouse

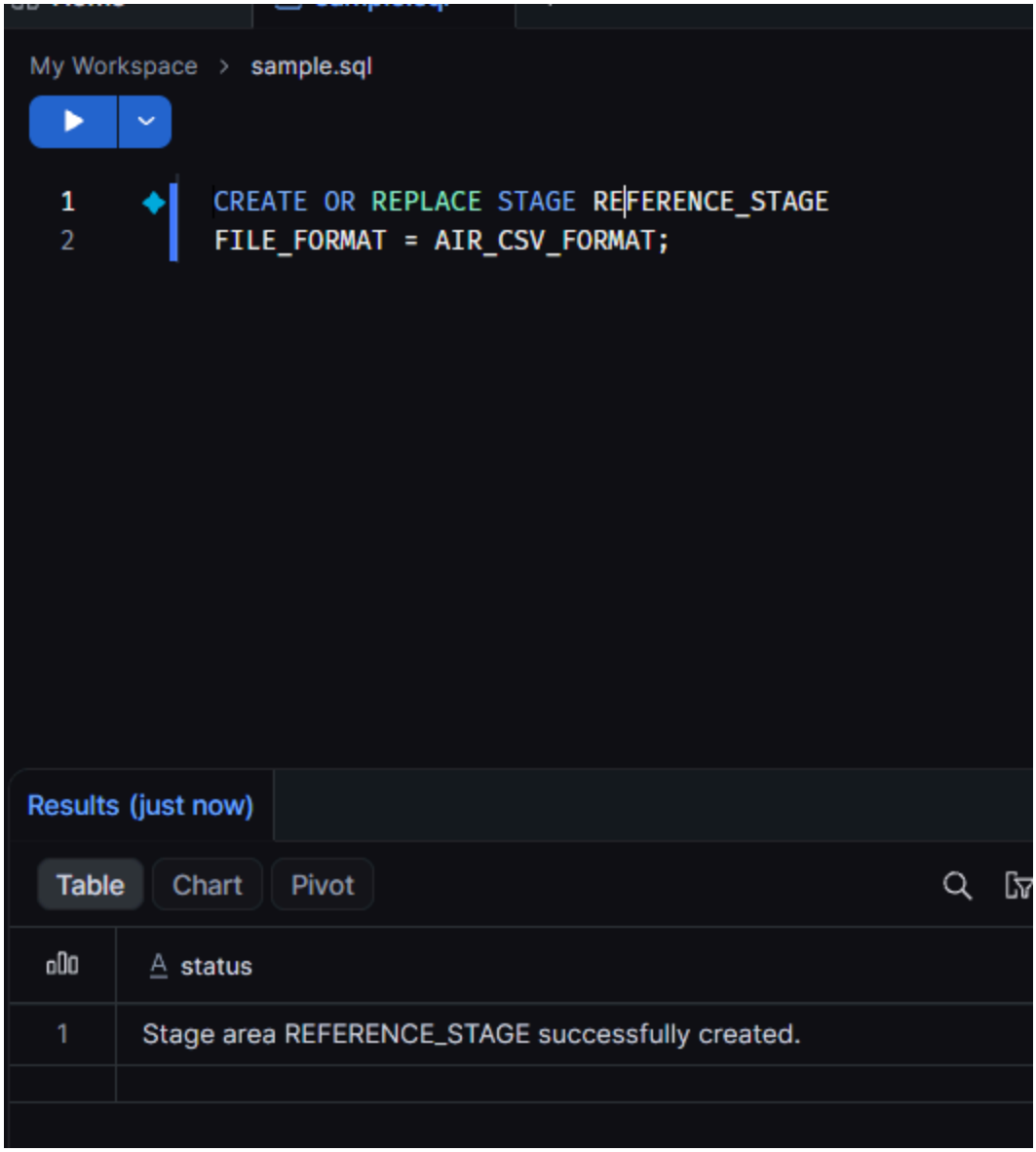


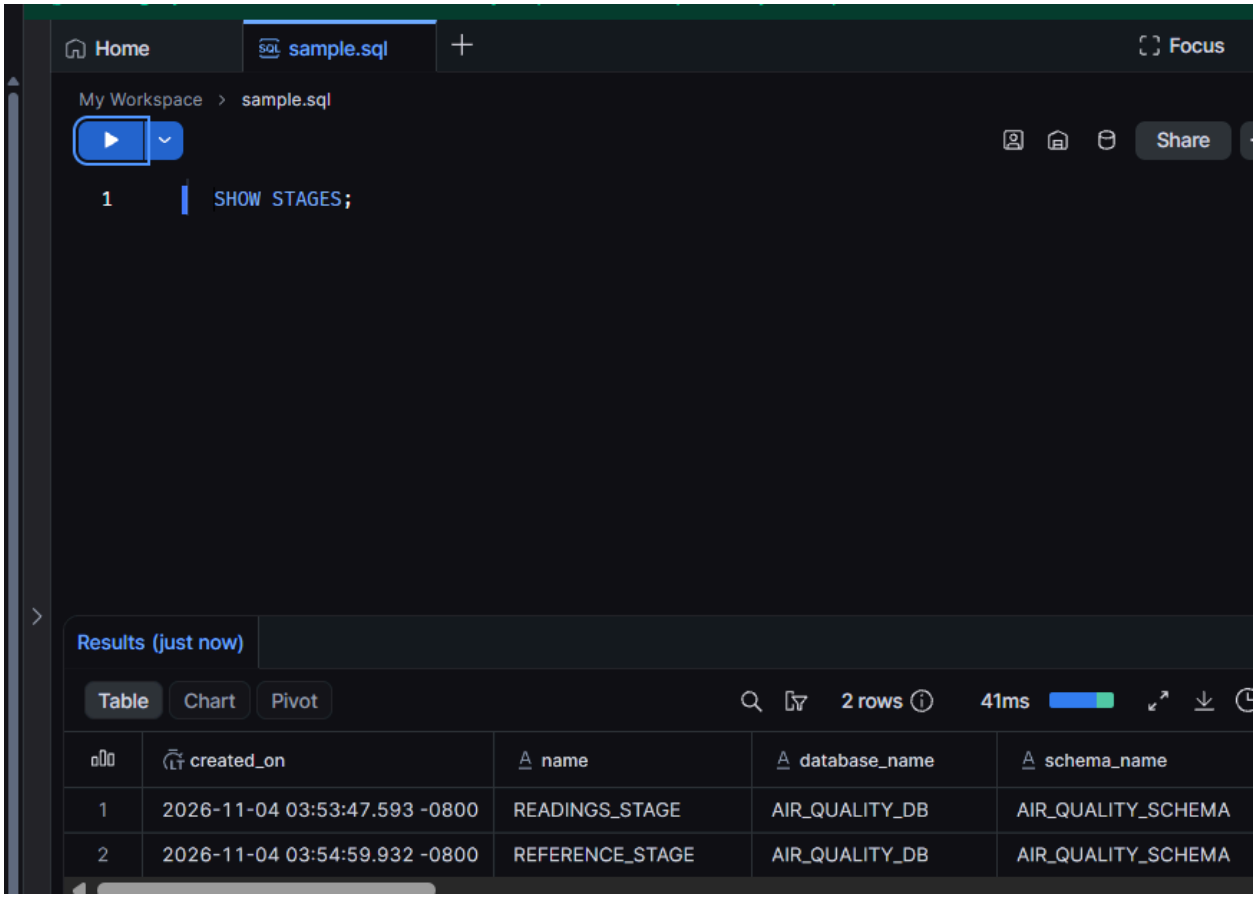
Create CSV file format



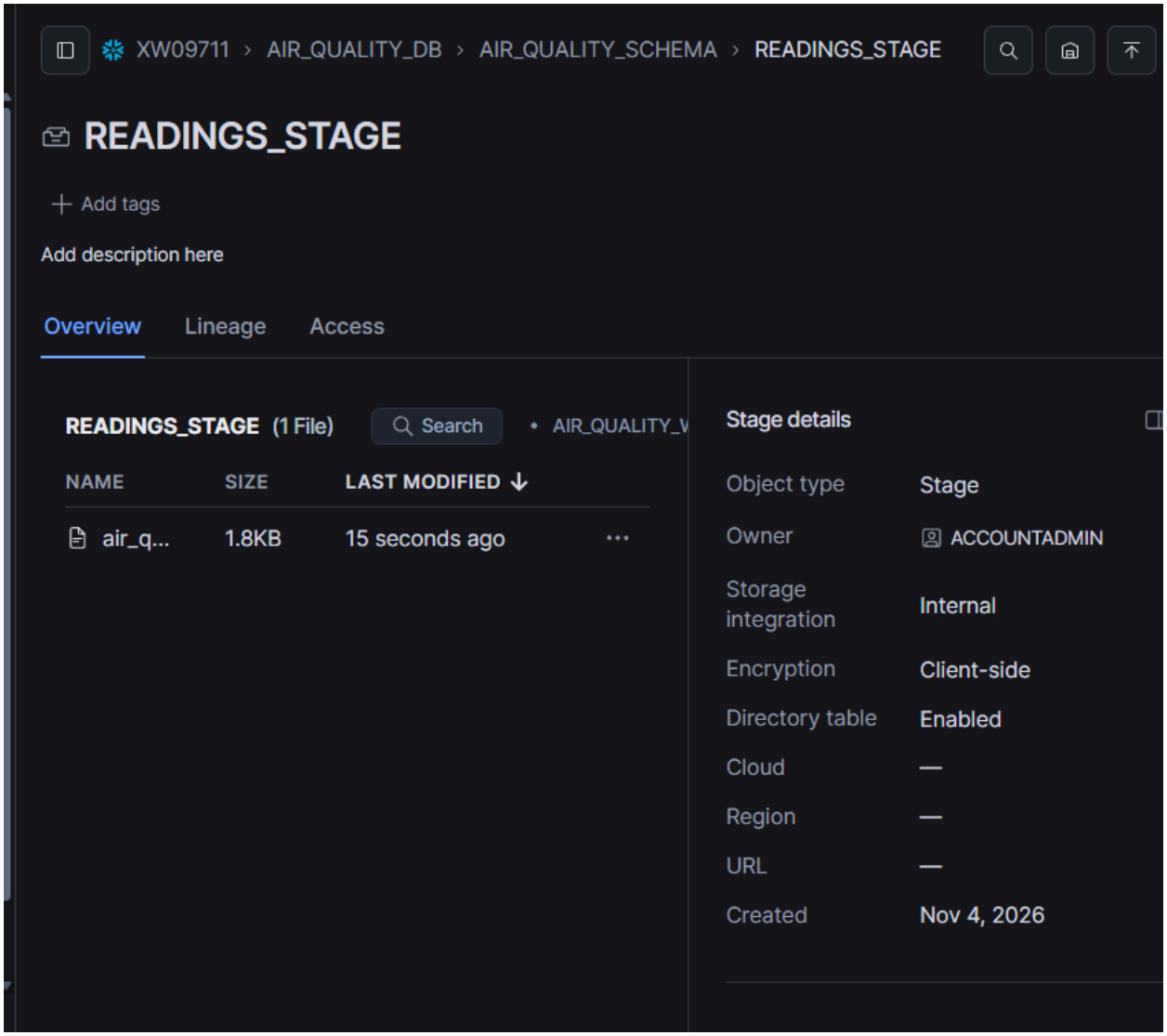

Home
sample.sql
Focus
My Workspace
sample.sql
Share
1
DESC FILE FORMAT AIR_CSV_FORMAT;
Results (just now)
Table
Chart
Pivot
24 rows
39ms
property
property_type
property_value
property_default
BINARY_FO... 4.2%
COMPRESS... 4.2%
+22 more
String 54.2%
Boolean 37.5%
+2 more
true 20.8%
AUTO 16.7%
+12 more
true 20.8%
AUTO 16.7%
+11 more
1 TYPE String CSV CSV
2 RECORD_DELIMITER String \n \n
3 FIELD_DELIMITER String , ,
4 FILE_EXTENSION String
5 SKIP_HEADER Integer 1 0
6 PARSE_HEADER Boolean false false

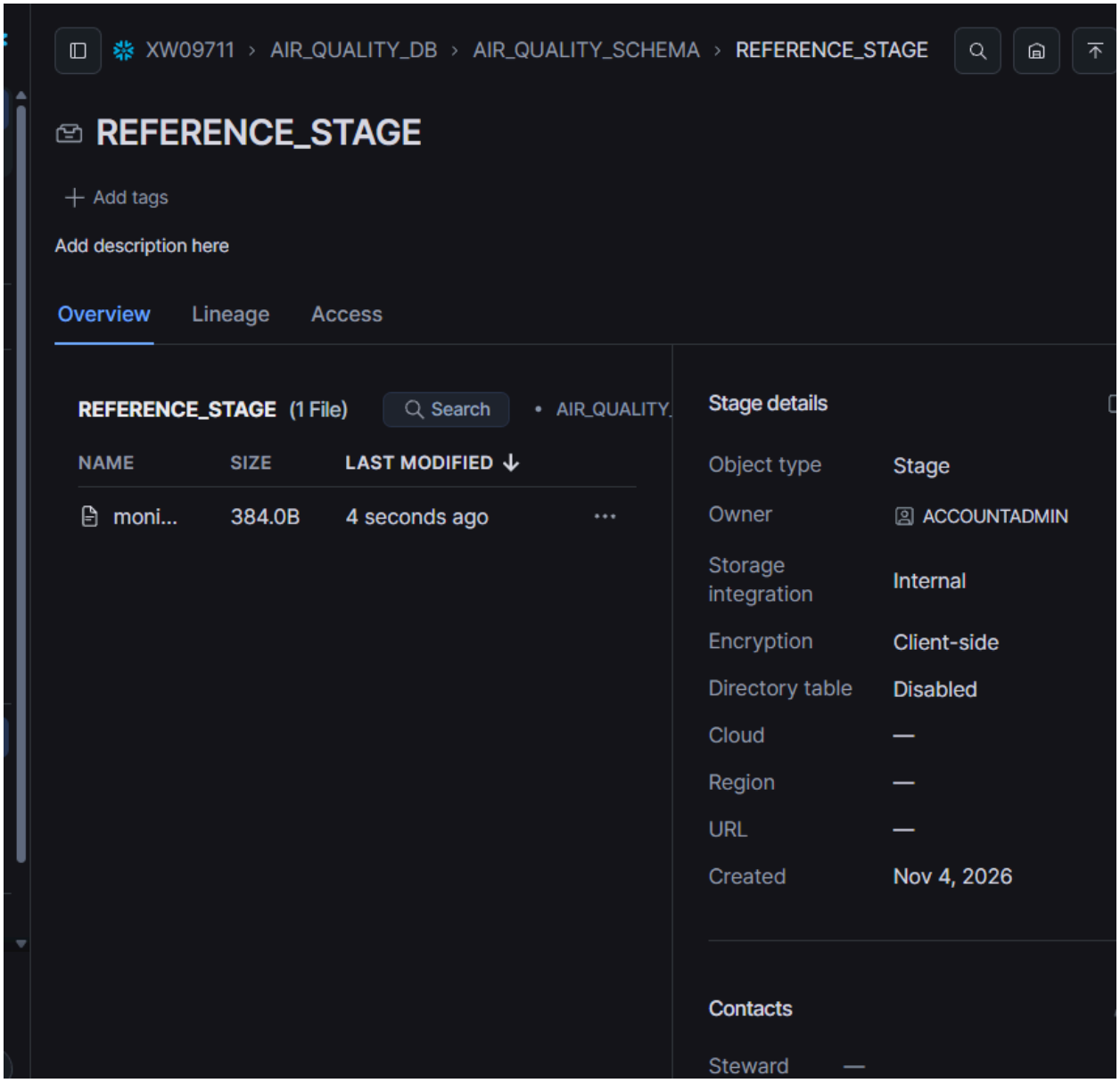

My Workspace
sample.sql
1 CREATE OR REPLACE STAGE READINGS_STAGE
2 FILE_FORMAT = AIR_CSV_FORMAT;
Results (just now)
Table
Chart
Pivot
1 row
77ms
status
1
Stage area READINGS_STAGE successfully created.

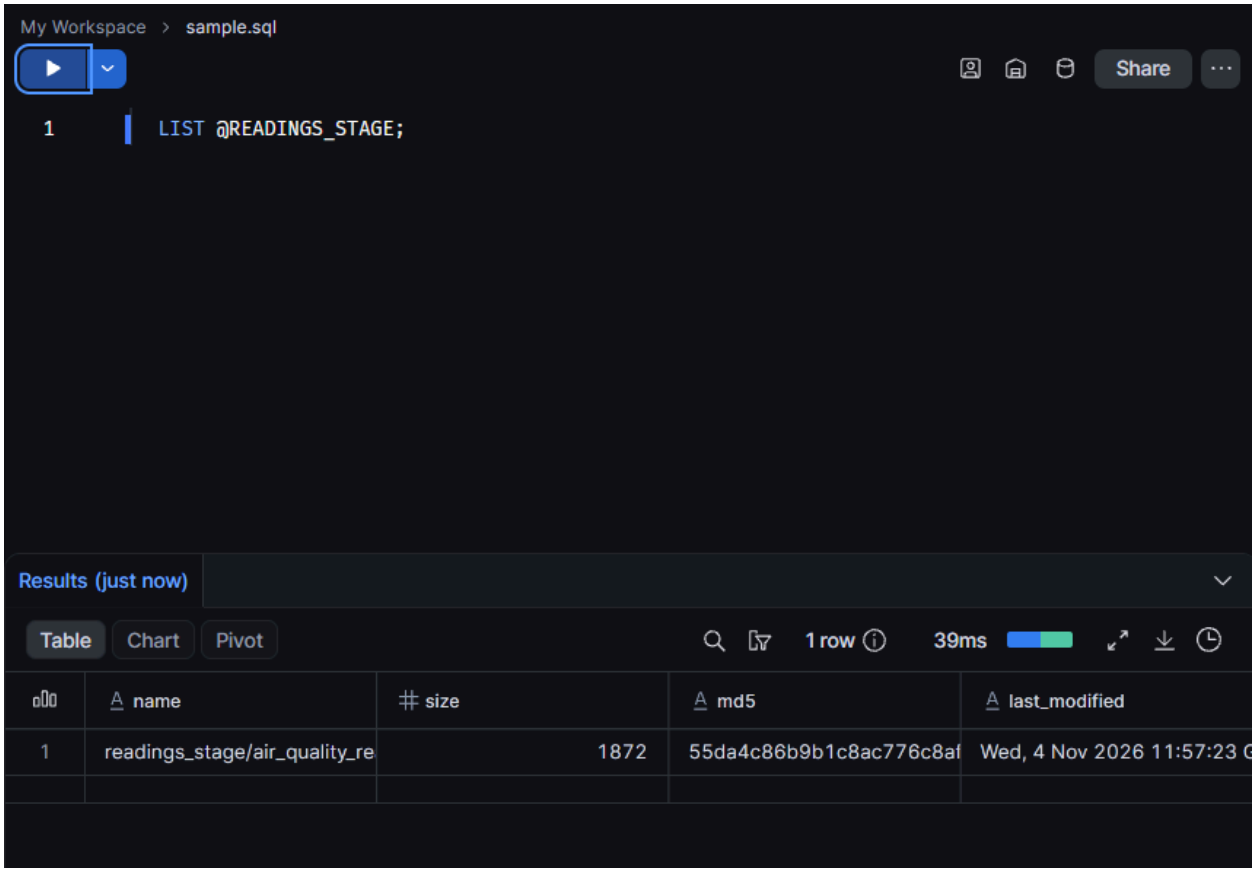

My Workspace
sample.sql
1 CREATE OR REPLACE STAGE REFERENCE_STAGE
2 FILE_FORMAT = AIR_CSV_FORMAT;
Results (just now)
Table
Chart
Pivot
status
1
Stage area REFERENCE_STAGE successfully created.

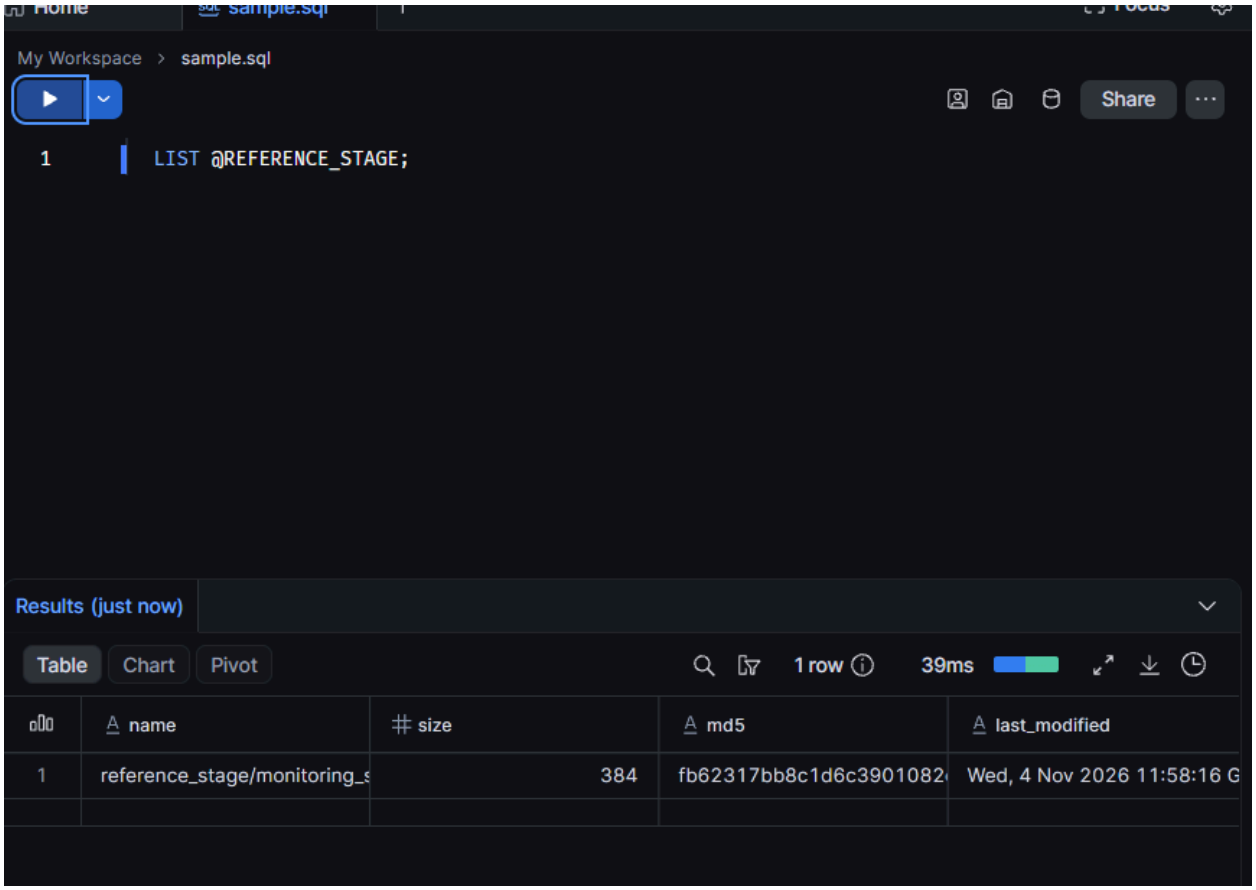

Home
sample.sql
Focus
My Workspace
sample.sql
Share
1
SHOW STAGES;
Results (just now)
Table
Chart
Pivot
2 rows
41ms
created_on
name
database_name
schema_name
1
2026-11-04 03:53:47.593 -0800
READINGS_STAGE
AIR_QUALITY_DB
AIR_QUALITY_SCHEMA
2
2026-11-04 03:54:59.932 -0800
REFERENCE_STAGE
AIR_QUALITY_DB
AIR_QUALITY_SCHEMA

Upload files to stage

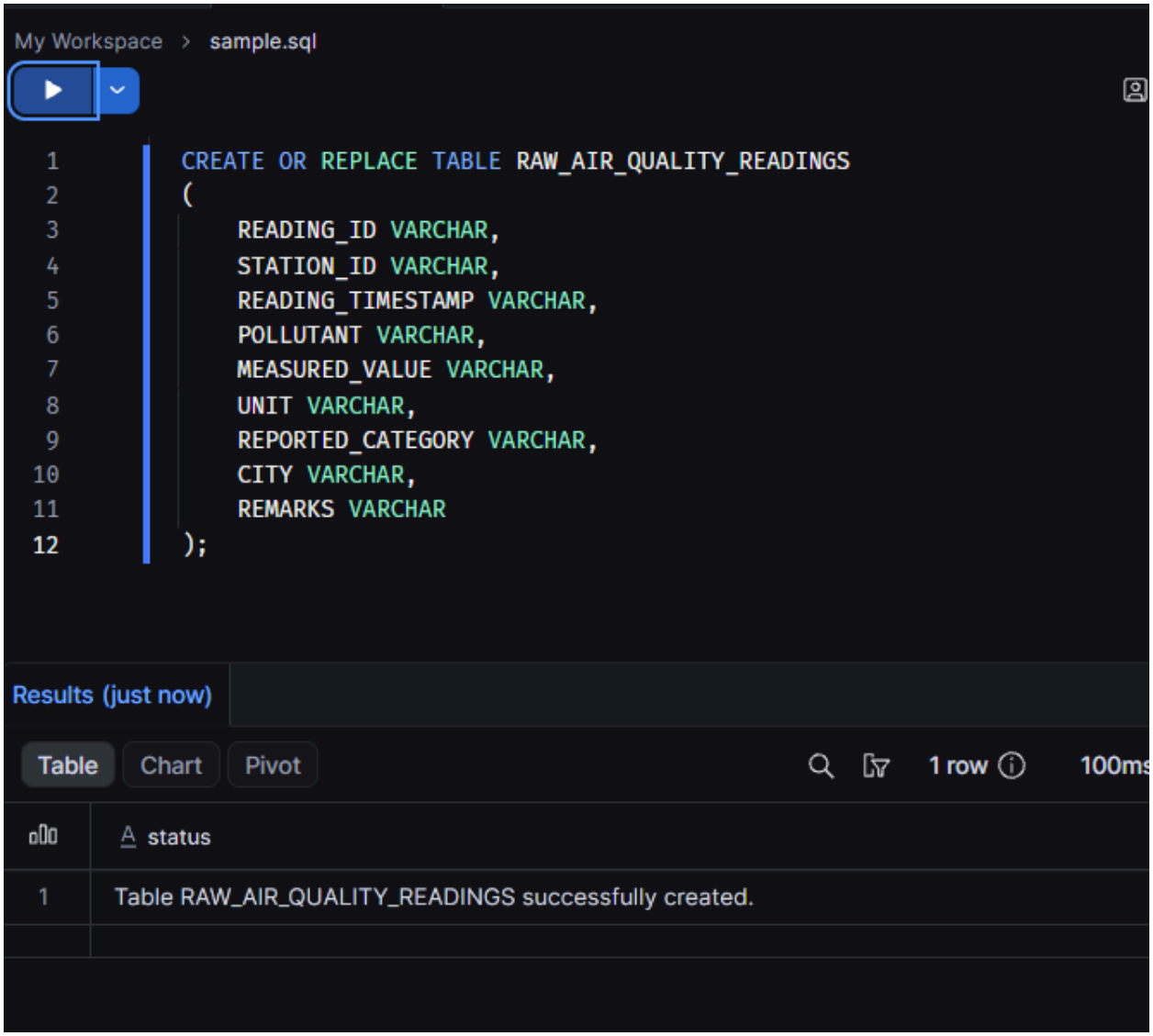


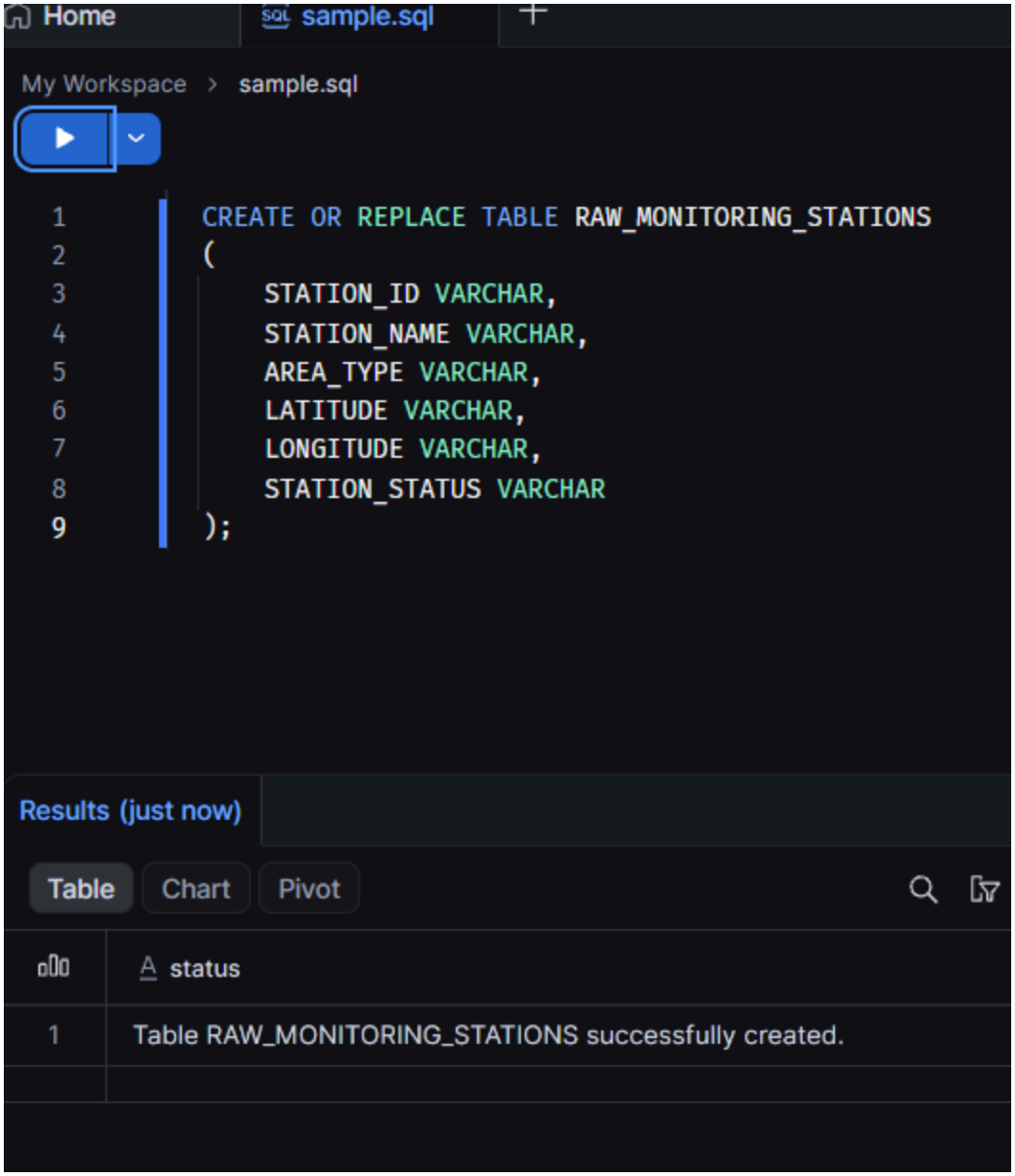

XW09711 › AIR_QUALITY_DB › AIR_QUALITY_SCHEMA › REFERENCE_STAGE
REFERENCE_STAGE
Add tags
Add description here
Overview
Lineage
Access
REFERENCE_STAGE (1 File)
Search
AIR_QUALITY
NAME
SIZE
LAST MODIFIED
moni...
384.0B
4 seconds ago
Stage details
Object type
Stage
Owner
ACCOUNTADMIN
Storage integration
Internal
Encryption
Client-side
Directory table
Disabled
Cloud
—
Region
—
URL
—
Created
Nov 4, 2026
Contacts
Steward
—


My Workspace
sample.sql
Share
1
LIST @READINGS_STAGE;
Results (just now)
Table
Chart
Pivot
1 row
39ms
name
size
md5
last_modified
1
readings_stage/air_quality_re
1872
55da4c86b9b1c8ac776c8af
Wed, 4 Nov 2026 11:57:23

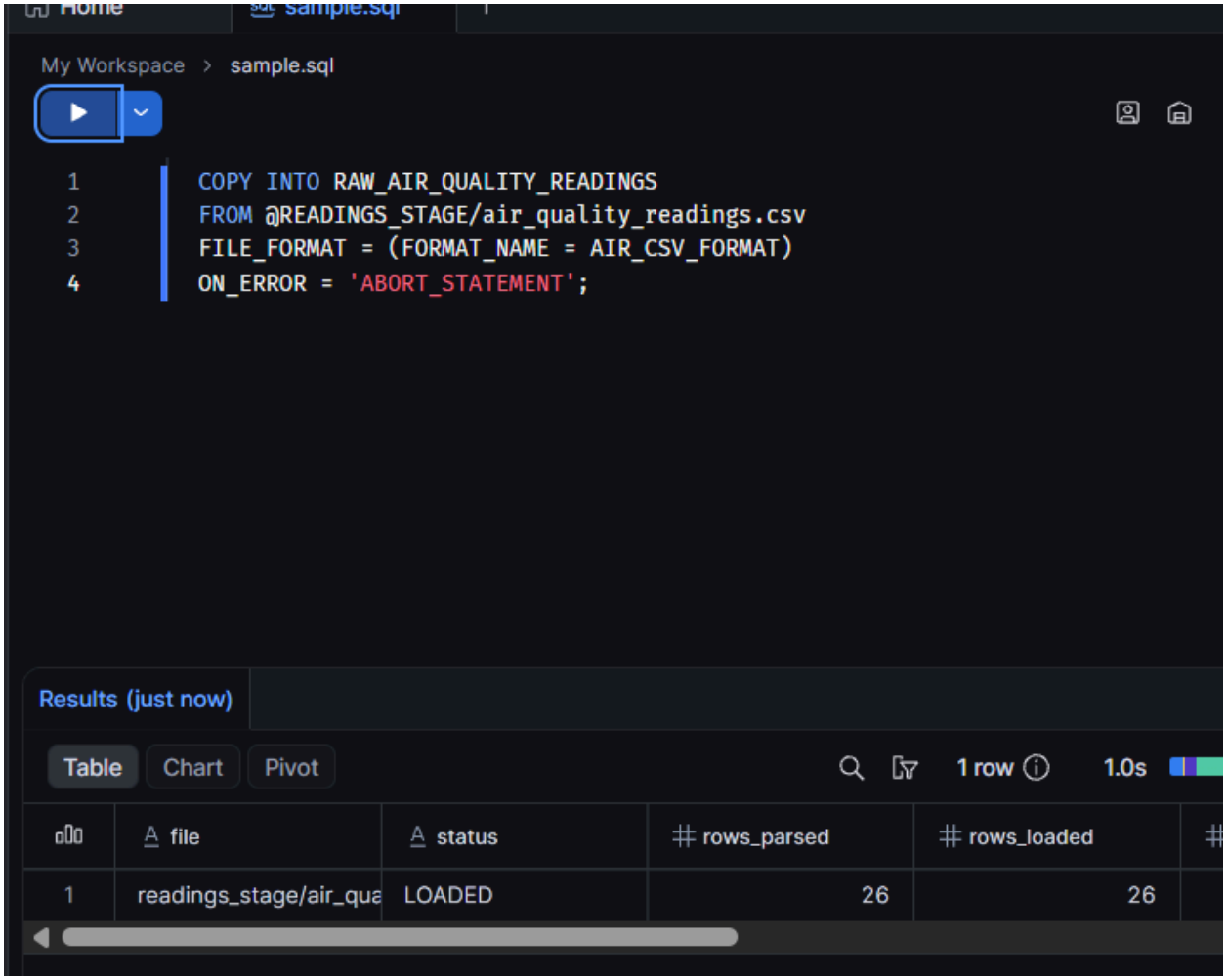

Home
sample.sql
My Workspace > sample.sql
Share
1 LIST @REFERENCE_STAGE;
Results (just now)
Table
Chart
Pivot
1 row
39ms
name
size
md5
last_modified
1
reference_stage/monitoring_s
384
fb62317bb8c1d6c3901082
Wed, 4 Nov 2026 11:58:16 G

Create table

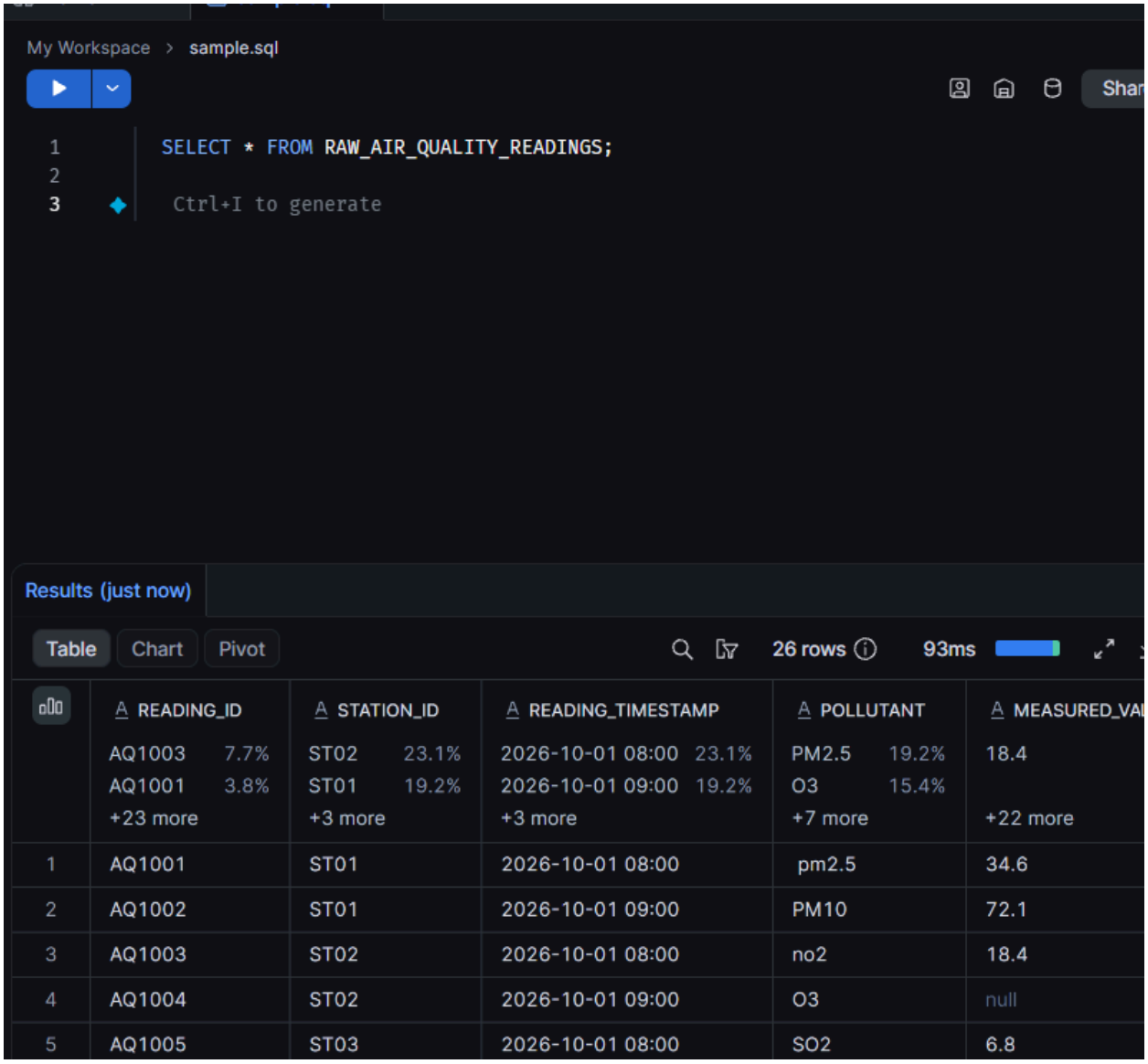



Home
sample.sql
My Workspace
sample.sql
CREATE OR REPLACE TABLE RAW_MONITORING_STATIONS
(
STATION_ID VARCHAR,
STATION_NAME VARCHAR,
AREA_TYPE VARCHAR,
LATITUDE VARCHAR,
LONGITUDE VARCHAR,
STATION_STATUS VARCHAR
);
Results (just now)
Table
Chart
Pivot
status
1
Table RAW_MONITORING_STATIONS successfully created.

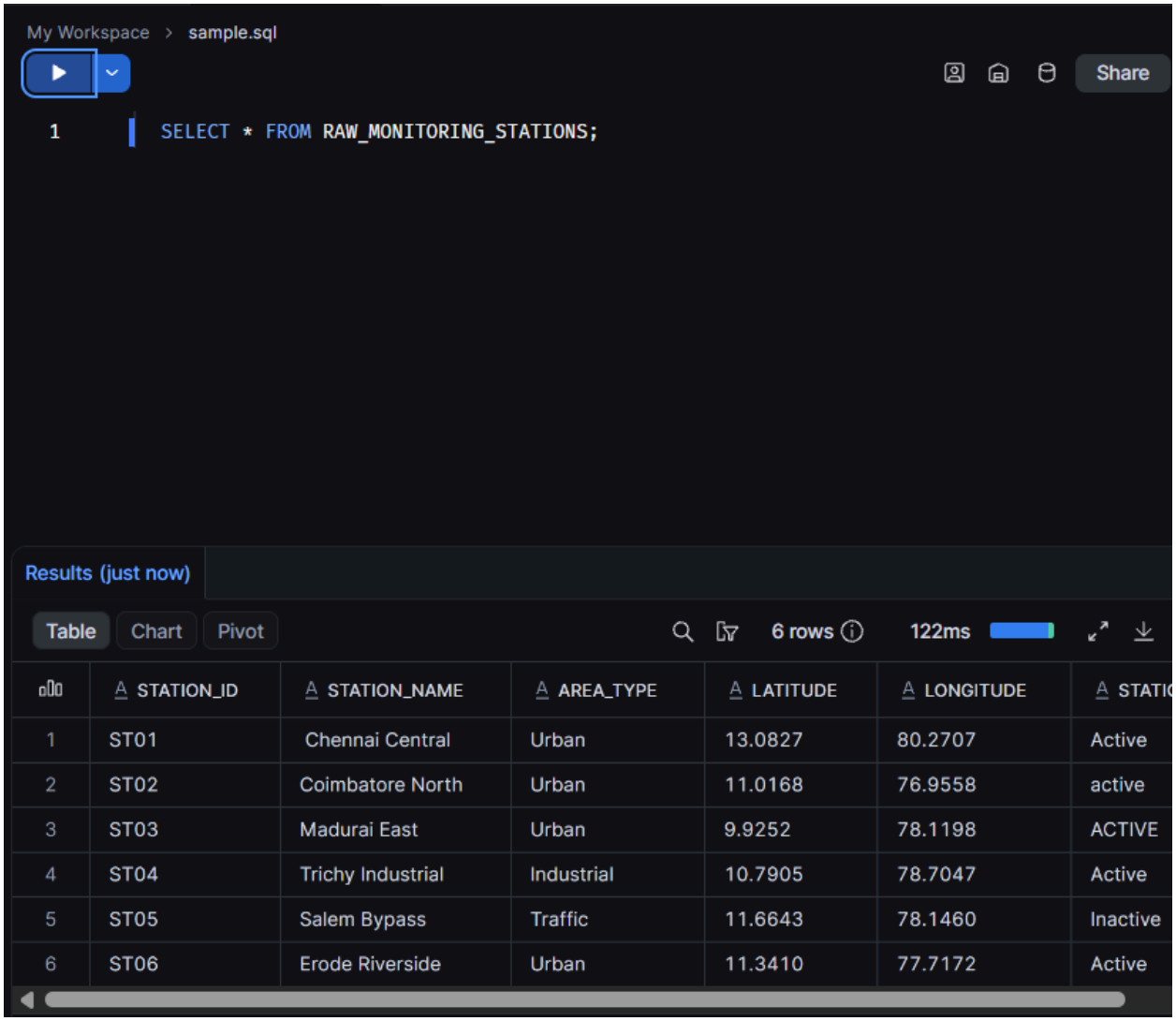
Load data to table

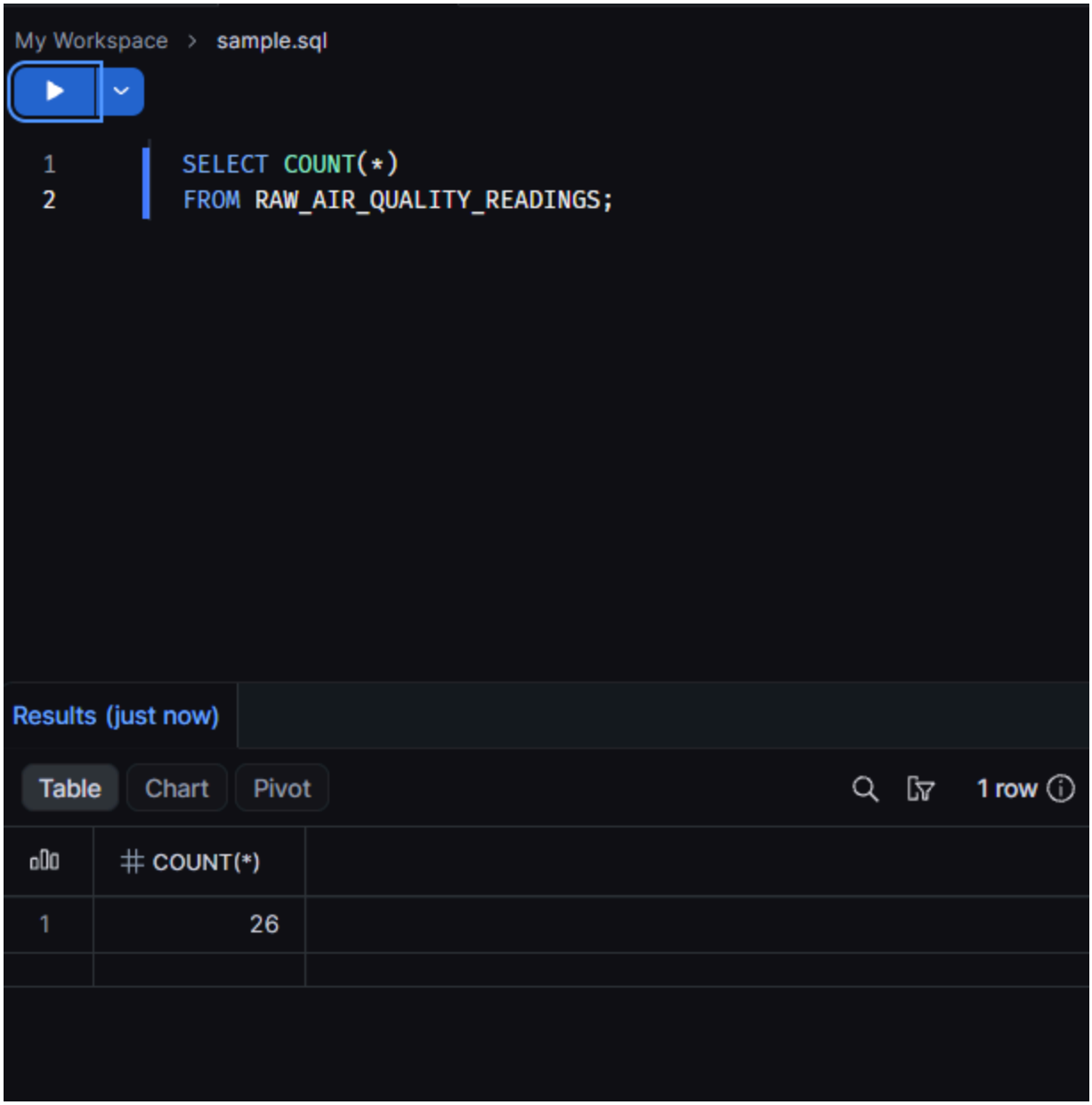


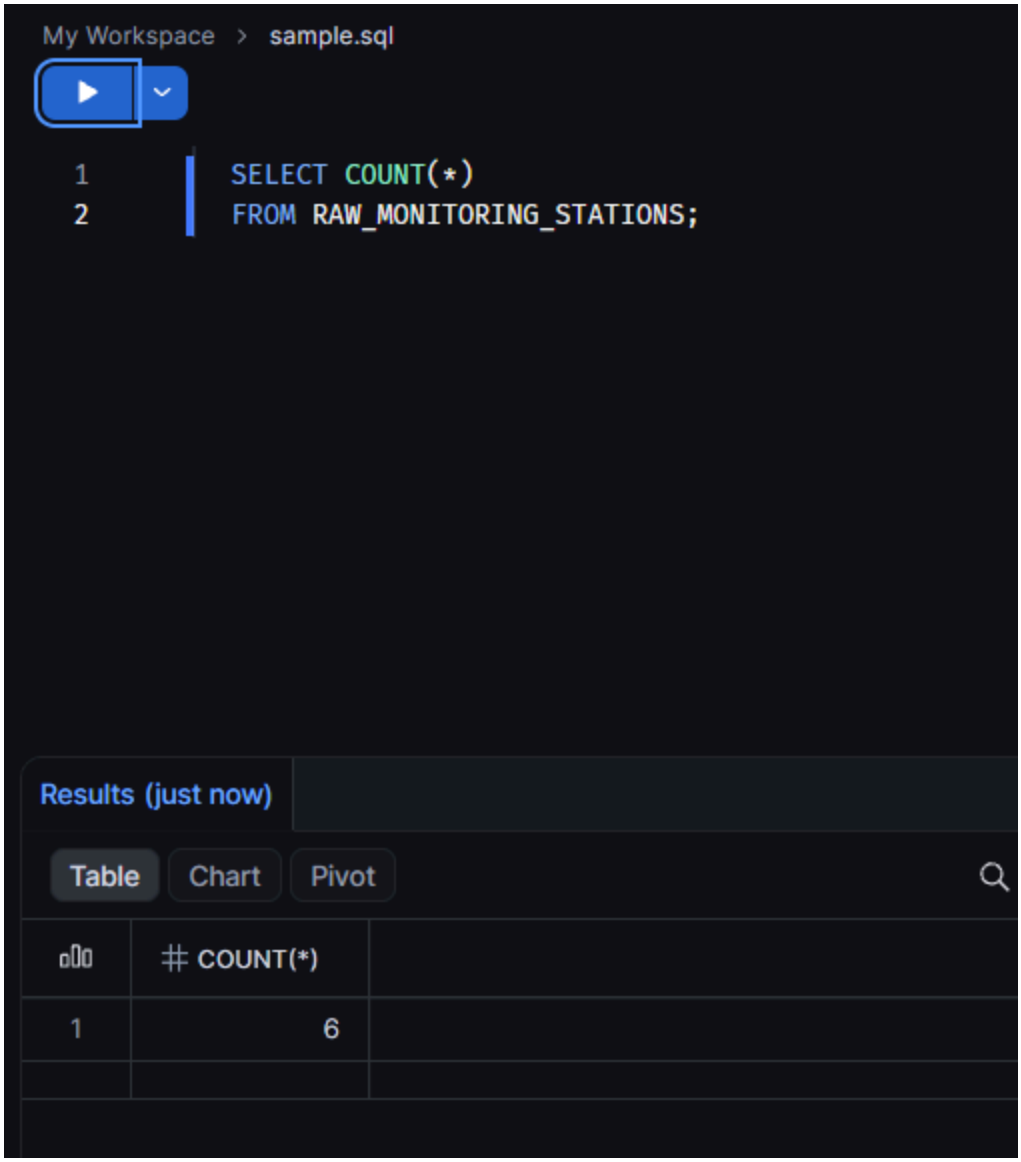
Validate RAW tables

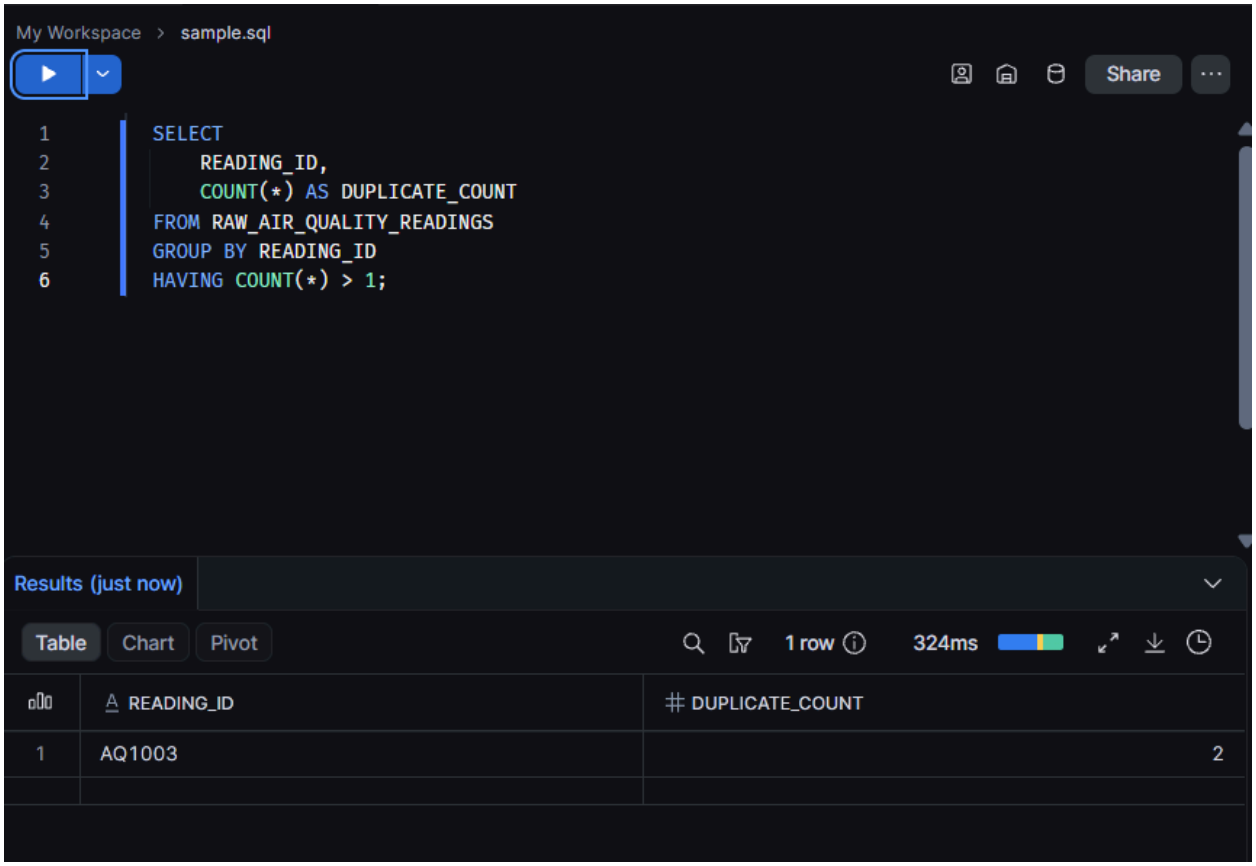


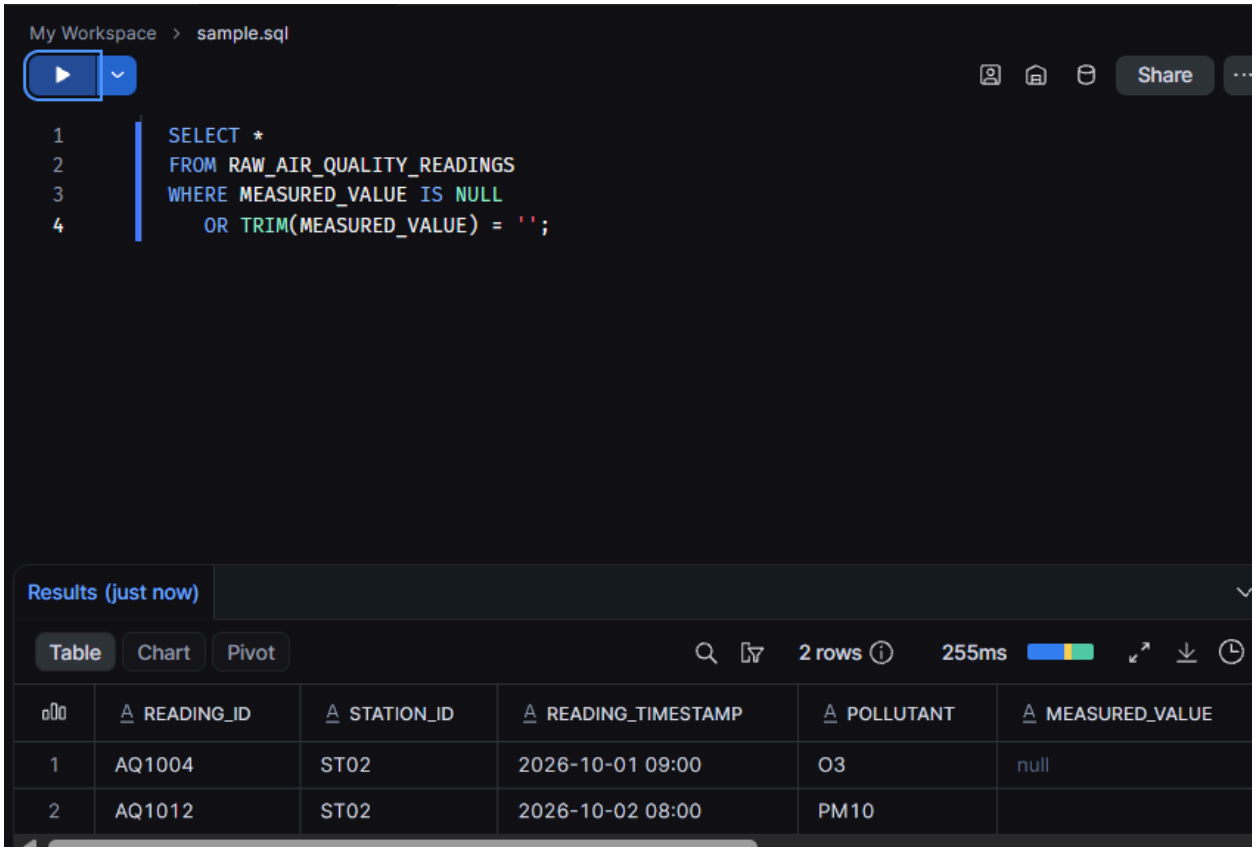
	READING_ID	STATION_ID	READING_TIMESTAMP	POLLUTANT	MEASURED_VAL
	AQ1003 7.7% AQ1001 3.8% +23 more	ST02 23.1% ST01 19.2% +3 more	2026-10-01 08:00 23.1% 2026-10-01 09:00 19.2% +3 more	PM2.5 19.2% O3 15.4% +7 more	18.4 +22 more
1	AQ1001	ST01	2026-10-01 08:00	pm2.5	34.6
2	AQ1002	ST01	2026-10-01 09:00	PM10	72.1
3	AQ1003	ST02	2026-10-01 08:00	no2	18.4
4	AQ1004	ST02	2026-10-01 09:00	O3	null
5	AQ1005	ST03	2026-10-01 08:00	SO2	6.8


My Workspace
sample.sql
Share
1
SELECT * FROM RAW_MONITORING_STATIONS;
Results (just now)
Table
Chart
Pivot
6 rows
122ms
STATION_ID
STATION_NAME
AREA_TYPE
LATITUDE
LONGITUDE
1 ST01 Chennai Central Urban 13.0827 80.2707 Active
2 ST02 Coimbatore North Urban 11.0168 76.9558 active
3 ST03 Madurai East Urban 9.9252 78.1198 ACTIVE
4 ST04 Trichy Industrial Industrial 10.7905 78.7047 Active
5 ST05 Salem Bypass Traffic 11.6643 78.1460 Inactive
6 ST06 Erode Riverside Urban 11.3410 77.7172 Active


My Workspace
sample.sql
1 SELECT COUNT(*)
2 FROM RAW_AIR_QUALITY_READINGS;
Results (just now)
Table
Chart
Pivot
1 row
COUNT(*)
1
26


My Workspace > sample.sql
1 SELECT COUNT(*)
2 FROM RAW_MONITORING_STATIONS;
Results (just now)
Table
Chart
Pivot
COUNT(*)
1
6


My Workspace > sample.sql
Share
1 SELECT
2 READING_ID,
3 COUNT(*) AS DUPLICATE_COUNT
4 FROM RAW_AIR_QUALITY_READINGS
5 GROUP BY READING_ID
6 HAVING COUNT(*) > 1;
Results (just now)
Table
Chart
Pivot
1 row
324ms
READING_ID
DUPLICATE_COUNT
1
AQ1003
2



Create CLEAN table for air quality readings

```
CREATE OR REPLACE TABLE CLEAN_AIR_QUALITY_READINGS AS

SELECT
   TRIM(READING_ID) AS READING_ID,

   TRIM(STATION_ID) AS STATION_ID,

   TRY_TO_TIMESTAMP_NTZ(
      TRIM(READING_TIMESTAMP)
   ) AS READING_TIMESTAMP,

   UPPER(TRIM(POLLUTANT)) AS POLLUTANT,

   CAST(
      NULLIF(TRIM(MEASURED_VALUE), '')
      AS NUMBER(10,2)
   ) AS MEASURED_VALUE,
```

```
    TRIM(UNIT) AS UNIT,

    CASE
        WHEN UPPER(TRIM(REPORTED_CATEGORY)) = 'GOOD'
            THEN 'GOOD'

        WHEN UPPER(TRIM(REPORTED_CATEGORY)) = 'MODERATE'
            THEN 'MODERATE'

        WHEN UPPER(TRIM(REPORTED_CATEGORY)) = 'POOR'
            THEN 'POOR'

        WHEN UPPER(TRIM(REPORTED_CATEGORY)) = 'UNHEALTHY'
            THEN 'UNHEALTHY'

        ELSE 'UNKNOWN'
    END AS AIR_QUALITY_CATEGORY,

    UPPER(TRIM(CITY)) AS CITY,

    COALESCE(
        NULLIF(TRIM(REMARKS), ''),
        'NO REMARKS'
    ) AS REMARKS,

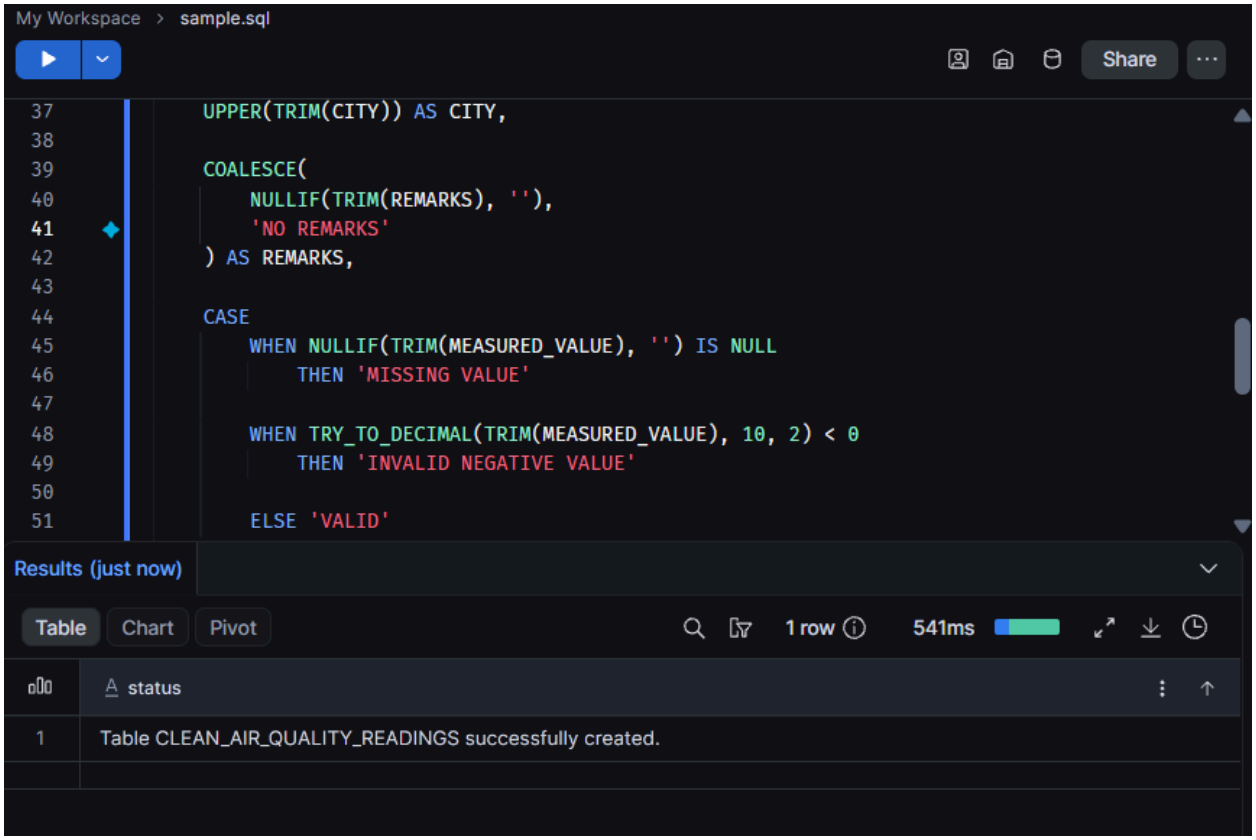
    CASE
        WHEN NULLIF(TRIM(MEASURED_VALUE), '') IS NULL
            THEN 'MISSING VALUE'

        WHEN TRY_TO_DECIMAL(TRIM(MEASURED_VALUE), 10, 2) < 0
            THEN 'INVALID NEGATIVE VALUE'

        ELSE 'VALID'
    END AS DATA_QUALITY_STATUS

FROM RAW_AIR_QUALITY_READINGS

QUALIFY ROW_NUMBER() OVER (
    PARTITION BY TRIM(READING_ID)
    ORDER BY TRIM(READING_ID)
) = 1;
```




```
CREATE OR REPLACE TABLE CLEAN_MONITORING_STATIONS AS

SELECT
   TRIM(STATION_ID) AS STATION_ID,

   INITCAP(TRIM(STATION_NAME)) AS STATION_NAME,

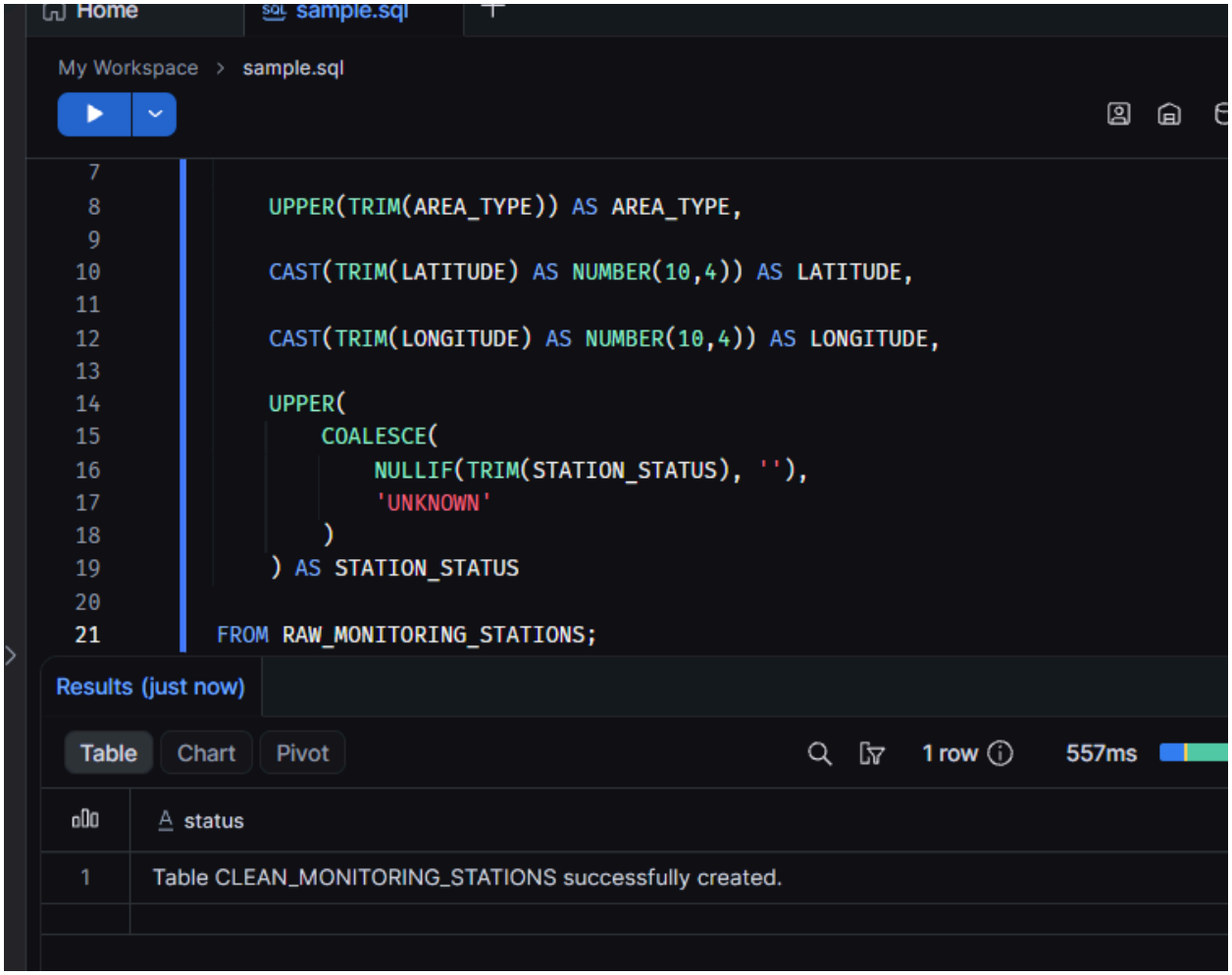
   UPPER(TRIM(AREA_TYPE)) AS AREA_TYPE,

   CAST(TRIM(LATITUDE) AS NUMBER(10,4)) AS LATITUDE,

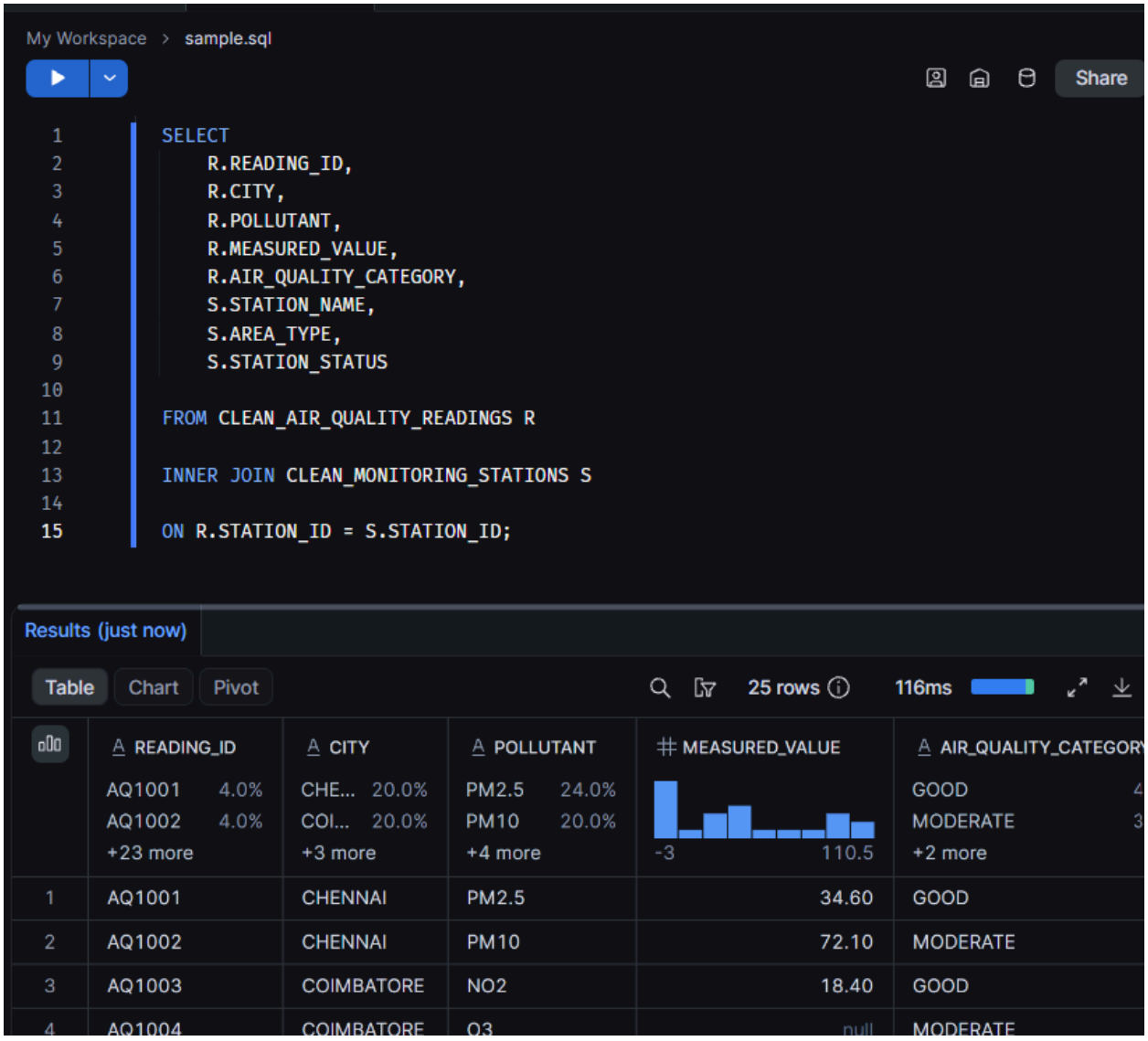
   CAST(TRIM(LONGITUDE) AS NUMBER(10,4)) AS LONGITUDE,

   UPPER(
      COALESCE(
         NULLIF(TRIM(STATION_STATUS), ''),
         'UNKNOWN'
      )
   ) AS STATION_STATUS

FROM RAW_MONITORING_STATIONS;
```


Home
sample.sql
My Workspace > sample.sql
7
8 UPPER(TRIM(AREA_TYPE)) AS AREA_TYPE,
9
10 CAST(TRIM(LATITUDE) AS NUMBER(10,4)) AS LATITUDE,
11
12 CAST(TRIM(LONGITUDE) AS NUMBER(10,4)) AS LONGITUDE,
13
14 UPPER(
15 COALESCE(
16 NULLIF(TRIM(STATION_STATUS), ''),
17 'UNKNOWN'
18)
19) AS STATION_STATUS
20
21 FROM RAW_MONITORING_STATIONS;
Results (just now)
Table
Chart
Pivot
1 row
557ms
status
1
Table CLEAN_MONITORING_STATIONS successfully created.

SQL Optimisation



My Workspace > sample.sql

Share

```sql
SELECT
    CITY,
    COUNT(*) AS TOTAL_READINGS,
    AVG(MEASURED_VALUE) AS AVG_MEASUREMENT

FROM CLEAN_AIR_QUALITY_READINGS

WHERE DATA_QUALITY_STATUS = 'VALID'

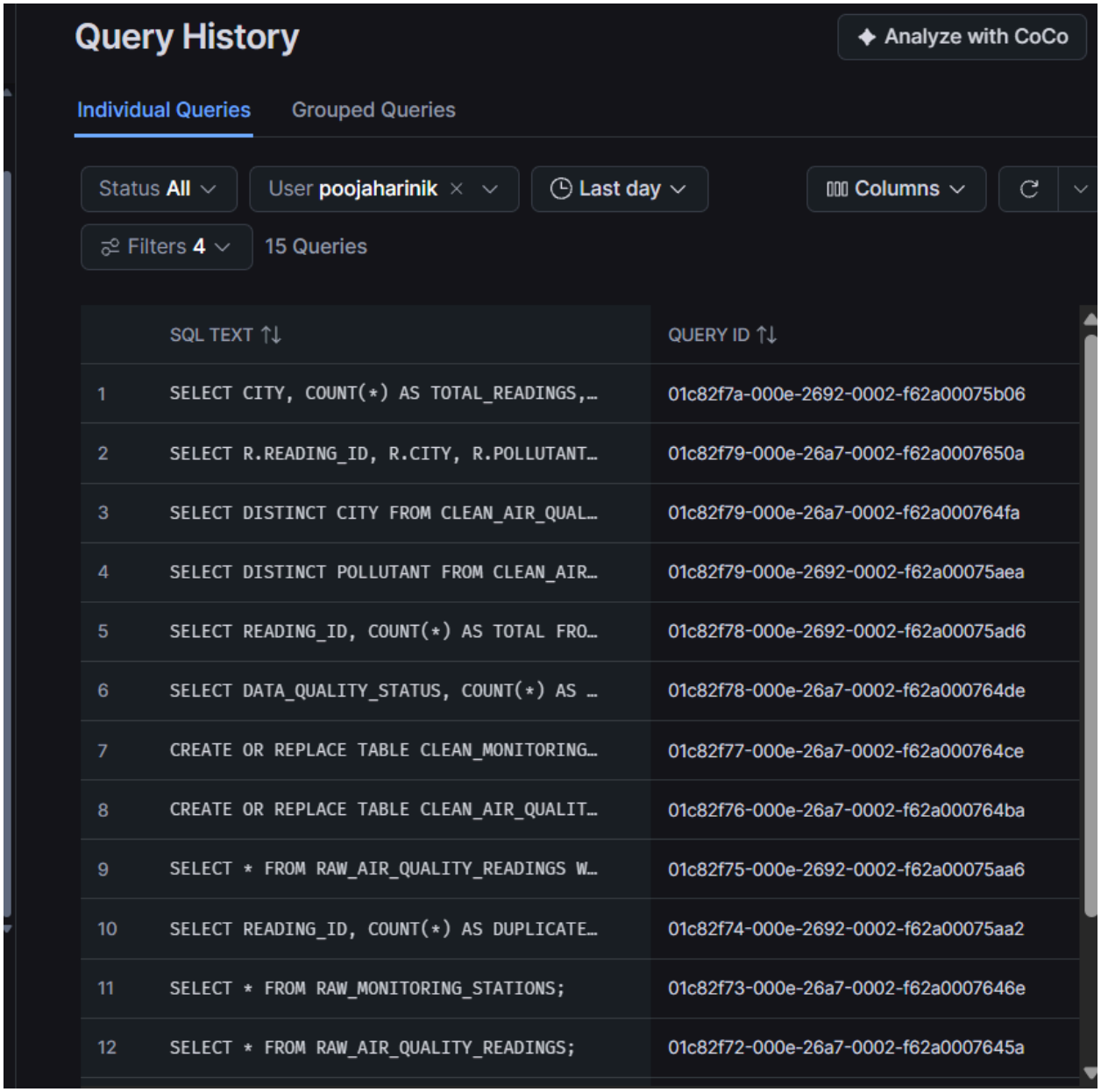
GROUP BY CITY
ORDER BY AVG_MEASUREMENT DESC;
```

Results (just now)

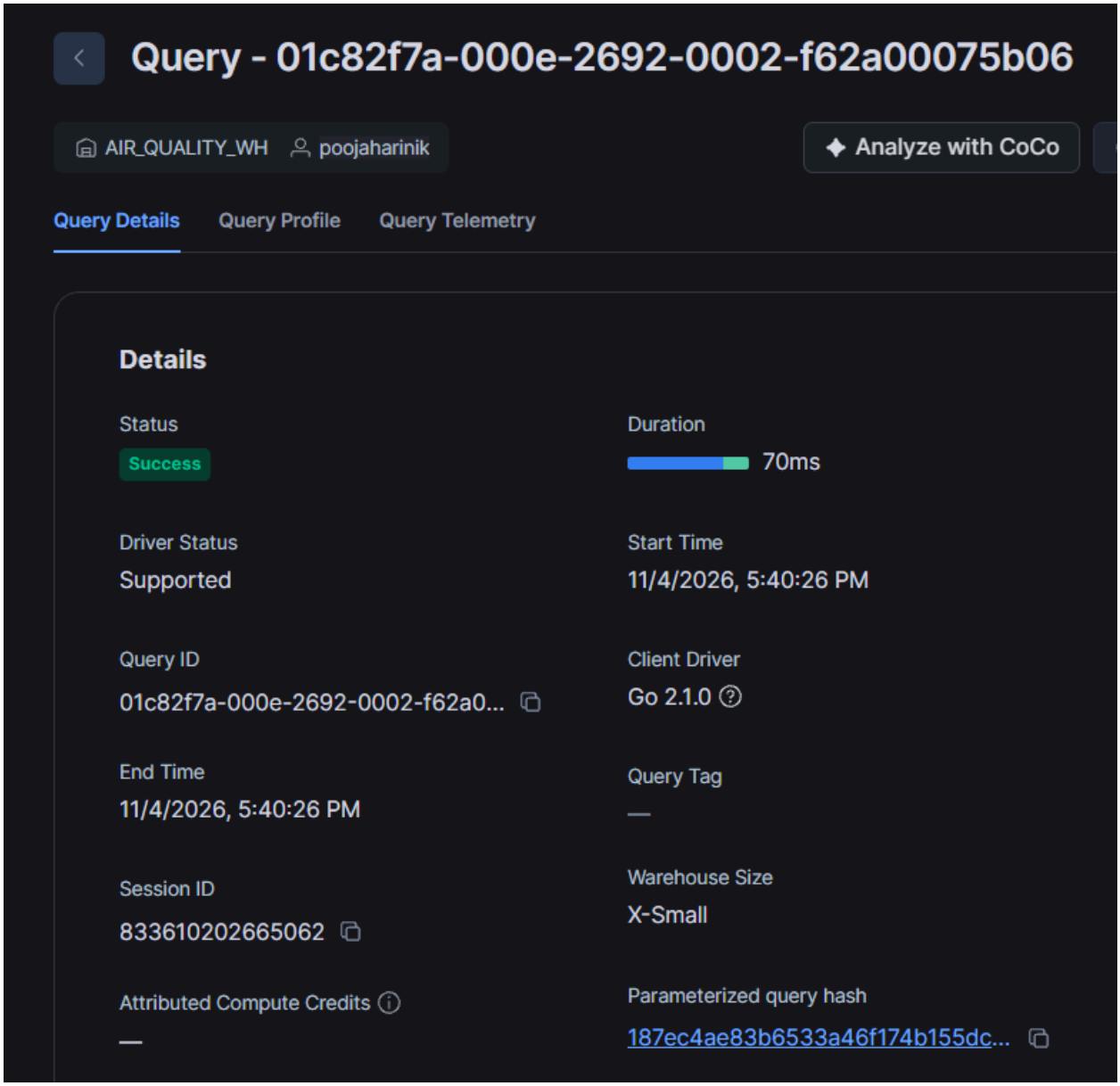
Table | Chart | Pivot

5 rows | 70ms

	CITY	TOTAL_READINGS	AVG_MEASUREMENT
1	TRICHY	5	66.7000
2	CHENNAI	5	43.3200
3	SALEM	4	39.2000
4	MADURAI	5	27.5000
5	COIMBATORE	3	25.2333



	SQL TEXT ↑↓	QUERY ID ↑↓
1	SELECT CITY, COUNT(*) AS TOTAL_READINGS,…	01c82f7a-000e-2692-0002-f62a00075b06
2	SELECT R.READING_ID, R.CITY, R.POLLUTANT…	01c82f79-000e-26a7-0002-f62a0007650a
3	SELECT DISTINCT CITY FROM CLEAN_AIR_QUAL…	01c82f79-000e-26a7-0002-f62a000764fa
4	SELECT DISTINCT POLLUTANT FROM CLEAN_AIR…	01c82f79-000e-2692-0002-f62a00075aea
5	SELECT READING_ID, COUNT(*) AS TOTAL FRO…	01c82f78-000e-2692-0002-f62a00075ad6
6	SELECT DATA_QUALITY_STATUS, COUNT(*) AS …	01c82f78-000e-26a7-0002-f62a000764de
7	CREATE OR REPLACE TABLE CLEAN_MONITORING…	01c82f77-000e-26a7-0002-f62a000764ce
8	CREATE OR REPLACE TABLE CLEAN_AIR_QUALIT…	01c82f76-000e-26a7-0002-f62a000764ba
9	SELECT * FROM RAW_AIR_QUALITY_READINGS W…	01c82f75-000e-2692-0002-f62a00075aa6
10	SELECT READING_ID, COUNT(*) AS DUPLICATE…	01c82f74-000e-2692-0002-f62a00075aa2
11	SELECT * FROM RAW_MONITORING_STATIONS;	01c82f73-000e-26a7-0002-f62a0007646e
12	SELECT * FROM RAW_AIR_QUALITY_READINGS;	01c82f72-000e-26a7-0002-f62a0007645a


Query - 01c82f7a-000e-2692-0002-f62a00075b06
AIR_QUALITY_WH
poojaharinik
Analyze with CoCo
Query Details
Query Profile
Query Telemetry
Details
Status
Success
Duration
70ms
Driver Status
Supported
Start Time
11/4/2026, 5:40:26 PM
Query ID
01c82f7a-000e-2692-0002-f62a0...
Client Driver
Go 2.1.0
End Time
11/4/2026, 5:40:26 PM
Query Tag
—
Session ID
833610202665062
Warehouse Size
X-Small
Attributed Compute Credits
—
Parameterized query hash
187ec4ae83b6533a46f174b155dc...